SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO2 - ASSESSMENT TOOL - 1

Scenario Based Question

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO2**	**Assessment**	**Assessment Tool 1: Scenario Based Question**
Weightage	**50% of CIA**	**Total Marks**	**100**
CO2 Statement	**Design effective test cases using black-box, white-box testing techniques and to control methods flow.**		
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Code of Conduct:

I,B.Thejaswini, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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6 A specialized HR payroll system calculates monthly bonuses based on years of service and performance ratings. Testers found that employees at exactly 5, 10, and 15 years were receiving in correct payouts. Critically analyze the failure and evaluate the current test design. Apply Boundary Value Analysis (BVA) and Equivalence Class Partitioning (ECP) to redesign the test suite. Justify why these techniques are superior to exploratory testing for this specific financial logic.

1. Introduction

The HR Payroll System is a financial application that calculates an employee's monthly bonus using two major inputs: years of service and performance rating. Because the result affects employee salary, even a small boundary defect can create an incorrect financial payout. In the given case, testers observed that employees with exactly 5, 10 and 15 years of service received incorrect bonuses.

This case strongly indicates a boundary-related defect. The values 5, 10 and 15 are likely decision thresholds at which the system changes the applicable bonus rule. If the implementation uses incorrect operators such as > instead of >=, or if the boundary values are assigned to the wrong equivalence class, exact threshold employees may receive the neighboring slab's bonus.

1.1 Problem Statement

The existing test suite did not adequately detect the incorrect payout at the exact service thresholds of 5, 10 and 15 years. The objective is to critically analyze the failure and redesign the test suite using Equivalence Class Partitioning (ECP) and Boundary Value Analysis (BVA).

1.2 Objectives

- Identify the likely cause and impact of the payroll failure.
- Analyze weaknesses in the existing test design.
- Partition inputs into valid and invalid equivalence classes.
- Apply BVA around 5, 10 and 15 years of service.
- Include performance rating classes and their combinations with service boundaries.
- Create traceable black-box test cases with expected results.
- Explain why systematic BVA/ECP is preferable to relying only on exploratory testing for this financial logic.

1.3 Scope

The scope covers the bonus-calculation decision logic. It includes years of service, performance rating, boundary conditions, invalid inputs, expected bonus slab selection, and integration of the calculated bonus into the payroll result. Salary tax, attendance, deductions and other payroll modules are outside the main scope unless they are affected by the bonus value.

2. Understanding the Failure

2.1 Failure Scenario

Assume the business rule defines service bands at 5, 10 and 15 years. For demonstration, the following bonus rates are used. These rates are test assumptions; the actual organization's approved payroll policy must be substituted in a real system.

Service band	Illustrative bonus rate
0 to less than 5 years	5%
5 to less than 10 years	8%
10 to less than 15 years	12%
15 years and above	15%

The critical point is not the particular percentage but the boundary behavior. The system must consistently classify exactly 5, exactly 10 and exactly 15 years according to the approved rule.

2.2 Why the Failure is Critical

- Financial impact: an incorrect percentage directly changes employee pay.
- Employee trust: payroll errors can reduce confidence in the HR system.
- Compliance risk: repeated incorrect payments can create audit and reconciliation problems.
- Defect propagation: one incorrect bonus value can flow into payslips, payroll reports and accounting records.
- Regression risk: changing a service-band rule may accidentally break adjacent bands.

2.3 Likely Technical Causes

- Wrong relational operator: using > instead of >= at a threshold.
- Off-by-one or off-by-precision error in service calculation.
- Incorrect ordering of conditional statements.
- Boundary value excluded from the intended equivalence class.
- Rounding or date-calculation issue when converting joining dates into completed years.
- Insufficient test data around threshold values.
- Incorrect mapping between service band and bonus-rate table.

2.4 Example of a Defective Decision

Suppose the intended condition is: if years_of_service >= 5, assign the second slab. A defective implementation might use years_of_service > 5. In that case, 5 years is incorrectly treated as belonging to the lower slab, while 5.1 years may enter the next slab. Similar mistakes can occur at 10 and 15 years.

3. Critical Evaluation of the Current Test Design

3.1 Typical Weak Test Design

A weak test suite may contain only representative middle values such as 2, 7, 12 and 20 years, together with a few performance ratings. Such tests can confirm that the general calculation works but may completely miss the exact decision thresholds.

Weak test	**Why it is insufficient**
2 years	Does not test the 5-year transition.
7 years	Inside a band; does not test 5 or 10.
12 years	Inside a band; does not test 10 or 15.
20 years	Tests an upper region but not the 15-year boundary.
Only average ratings	May miss rating boundaries and invalid ratings.
Random employee data	May rarely or never produce exact thresholds.

3.2 Main Deficiencies

- No explicit boundary-focused tests for 5, 10 and 15.
- No immediate predecessor and successor tests such as 4/5/6, 9/10/11 and 14/15/16.
- Insufficient negative testing for invalid service and performance inputs.
- No clear equivalence classes.
- No traceability between requirements and test cases.
- Potentially inadequate combinations of service boundaries with performance ratings.
- No explicit check that the calculated bonus rate is mapped to the correct payroll amount.

3.3 Risk of Random Testing

Random values can produce valid coverage of broad regions but have a low probability of selecting every important decision boundary, especially when the input domain is large or continuous. A financial defect concentrated at exact thresholds therefore requires deliberate test design rather than depending on chance.

3.4 Improved Testing Strategy

The redesigned strategy uses ECP first to reduce the input space into meaningful classes. BVA is then applied around each critical service boundary. Performance rating boundaries are also tested. Finally, selected combinations verify that service and rating rules interact correctly.

4. Equivalence Class Partitioning (ECP)

Equivalence Class Partitioning divides the input domain into groups in which the system is expected to behave similarly. Instead of testing every possible input, at least one representative from each meaningful class is selected.

4.1 Service-Year Equivalence Classes

Class ID	Condition	Type	Representative
S1	Service < 0 years	Invalid	-1
S2	0 <= service < 5	Valid	2
S3	Service = 5	Valid boundary	5
S4	5 < service < 10	Valid	7
S5	Service = 10	Valid boundary	10
S6	10 < service < 15	Valid	12
S7	Service = 15	Valid boundary	15
S8	Service > 15	Valid	20

4.2 Performance Rating Equivalence Classes

For illustration, assume the approved rating scale is 1 to 5, where higher ratings qualify for higher bonus treatment. The exact business rule must be confirmed before execution.

Class ID	Condition	Type	Representative
P1	Rating < 1	Invalid	0
P2	1 <= rating < 3	Valid	2
P3	Rating = 3	Valid boundary	3
P4	3 < rating < 5	Valid	4
P5	Rating = 5	Valid boundary	5
P6	Rating > 5	Invalid	6

4.3 ECP Advantages in This Case

- Reduces the number of tests while retaining meaningful input classes.
- Ensures invalid inputs are explicitly considered.
- Separates normal regions from threshold-specific behavior.
- Provides a structured basis for test coverage and traceability.
- Helps identify missing classes before detailed test execution.

5. Boundary Value Analysis (BVA)

Boundary Value Analysis focuses on values at and immediately around the edges of equivalence classes. It is particularly suitable because the reported failures occur exactly at 5, 10 and 15 years.

5.1 Standard Boundary Set

For each critical boundary x, a practical BVA set is x-1, x, and x+1. When the system accepts fractional or date-derived service values, values immediately below and above the threshold should also be represented at the system's supported precision.

Boundary	Below	Exact	Above
5 years	4	5	6
10 years	9	10	11
15 years	14	15	16

5.2 Extended Precision Check

If the payroll system calculates service from joining dates rather than a manually entered integer, additional tests should use values such as 4.99, 5.00 and 5.01 years, or dates one day before, on, and one day after the anniversary. This is important because a system can appear correct for integer inputs but still fail when real employee dates are converted to completed service.

5.3 Performance Rating Boundaries

Rating boundary	Below	Exact	Above
Lower valid limit 1	0	1	2
Important midpoint 3	2	3	4
Upper valid limit 5	4	5	6

5.4 Why BVA Fits the Defect

The observed failures are concentrated at decision points. BVA directly targets those points and their neighboring values. If 5 is wrong but 4 and 6 are correct, the test results strongly isolate the defect to the boundary condition rather than the general bonus calculation.

6. Redesigned Black-Box Test Suite

The following test cases use the illustrative business rule. Expected results should be replaced by the organization's approved bonus policy before execution.

TC ID	Service	Rating	Test purpose	Expected class/result
TC01	2	3	Normal lower band	S2; lower-band bonus
TC02	4	3	Below 5 boundary	S2; lower-band bonus
TC03	5	3	Exact 5 boundary	S3; 5-year slab
TC04	6	3	Above 5 boundary	S4; next slab
TC05	9	3	Below 10 boundary	S4; 5–10 slab
TC06	10	3	Exact 10 boundary	S5; 10-year slab
TC07	11	3	Above 10 boundary	S6; 10–15 slab
TC08	14	3	Below 15 boundary	S6; 10–15 slab
TC09	15	3	Exact 15 boundary	S7; 15+ slab
TC10	16	3	Above 15 boundary	S8; 15+ slab
TC11	20	3	Upper valid region	S8; 15+ slab
TC12	-1	3	Invalid service	Reject with validation message
TC13	2	0	Invalid rating	Reject with validation message
TC14	2	6	Rating above maximum	Reject with validation message
TC15	2	1	Lower valid rating	Accept and calculate
TC16	2	5	Upper valid rating	Accept and calculate

6.1 Boundary-Focused Regression Set

TC ID	Service values	Rating	Main check
B01	4, 5, 6	3	5-year transition
B02	9, 10, 11	3	10-year transition
B03	14, 15, 16	3	15-year transition
B04	4.99, 5.00, 5.01	3	Precision around 5
B05	9.99, 10.00, 10.01	3	Precision around 10
B06	14.99, 15.00, 15.01	3	Precision around 15

The boundary-focused set should be retained as a mandatory regression suite whenever the bonus calculation logic, service-date calculation, rate table, or payroll integration is changed.

7. Detailed Test Case Design

7.1 Test Case Template

Each test case should contain: Test Case ID, objective, precondition, input data, execution steps, expected result, actual result, pass/fail status, defect ID and remarks.

7.2 Detailed Boundary Tests

ID	Precondition	Input	Expected behavior
TC03	Employee record exists	Service=5; Rating=3	System must classify employee in the approved 5-year slab and calculate the corresponding bonus.
TC04	Employee record exists	Service=6; Rating=3	System must classify employee in the band immediately above 5 and use its approved rate.
TC06	Employee record exists	Service=10; Rating=3	System must classify employee in the approved 10-year slab.
TC07	Employee record exists	Service=11; Rating=3	System must classify employee in the band immediately above 10.
TC09	Employee record exists	Service=15; Rating=3	System must classify employee in the approved 15-year slab.
TC10	Employee record exists	Service=16; Rating=3	System must classify employee in the band immediately above 15.

7.3 Performance and Service Combination Tests

ID	Service	Rating	Purpose
C01	5	1	Lowest valid rating at service boundary
C02	5	5	Highest valid rating at service boundary
C03	10	1	Lowest valid rating at service boundary
C04	10	5	Highest valid rating at service boundary
C05	15	1	Lowest valid rating at service boundary
C06	15	5	Highest valid rating at service boundary
C07	4	5	Below service boundary with high rating
C08	16	1	Above service boundary with low rating

These combinations help distinguish a service-boundary defect from a performance-rating defect. If only exact service values fail across multiple ratings, the evidence points strongly to service-band classification.

8. Defect Report and Root-Cause Analysis

8.1 Sample Defect Report

Field	**Details**
Defect ID	PAY-BONUS-001
Title	Incorrect bonus payout at exactly 5 years of service
Severity	High
Priority	High
Module	Monthly Bonus Calculation
Input	Service = 5 years; valid performance rating
Expected	Employee receives the bonus defined for the 5-year slab
Actual	Employee receives the neighboring slab's bonus
Reproducibility	Consistent when service is exactly 5 years
Likely cause	Incorrect boundary operator/classification
Detection technique	BVA + ECP
Business impact	Incorrect employee compensation and payroll reporting
Suggested fix	Review threshold condition and service-date calculation; add boundary regression tests

8.2 Root-Cause Logic

A likely implementation defect is an incorrect comparison. For example, if the intended rule is service >= 5 but the program checks service > 5, exactly 5 years is incorrectly classified. The same pattern may occur at 10 and 15 years. Another possibility is incorrect ordering, where a broad condition captures the boundary before the more specific condition is evaluated.

8.3 White-Box Complement

Although the requested redesign is primarily black-box, CO2 also mentions white-box and control-flow methods. White-box testing can inspect the decision structure and ensure every branch of the bonus calculation is executed. A control-flow graph would represent the service and rating decisions as

nodes and branches. Branch coverage should include True and False outcomes for comparisons at 5, 10 and 15 years.

8.4 Example Control-Flow Checks

- Check the branch for service < 5.
- Check the branch for service >= 5 and < 10.
- Check the branch for service >= 10 and < 15.
- Check the branch for service >= 15.
- Check valid and invalid performance-rating branches.
- Verify every decision outcome is exercised by at least one test case.

9. Why BVA and ECP Are Superior to Exploratory Testing for This Logic

Exploratory testing is useful when requirements are unclear, when testers need to learn the product, and when unexpected behavior is likely. However, for a defined financial rule with explicit thresholds, systematic black-box techniques provide stronger and more measurable coverage.

Factor	**BVA + ECP**	**Exploratory testing alone**
Boundary coverage	Explicitly targets 5, 10, 15 and neighbors	May miss exact thresholds
Repeatability	Highly repeatable	Depends on tester path and experience
Traceability	Inputs map directly to requirements/classes	Coverage can be difficult to prove
Financial risk	Systematically checks high-risk transitions	May discover defects but does not guarantee boundaries
Regression	Compact boundary suite can be rerun	Exploratory sessions may vary
Invalid inputs	ECP explicitly defines invalid classes	Depends on tester creativity
Coverage evidence	Can document class and boundary coverage	Less objective unless carefully chartered

9.1 Critical Evaluation

BVA and ECP should not be described as universally superior to exploratory testing. The stronger conclusion is that they are superior as the primary technique for this specific defect pattern. Exploratory testing can still be used as a complementary technique to discover unexpected issues such as unusual employee records, date conversions, rounding, data migration problems, or integration defects.

10. Final Test Strategy

- Step 1: Confirm the approved bonus policy and exact boundary definitions.
- Step 2: Build ECP classes for service years and performance ratings.
- Step 3: Apply BVA around 5, 10 and 15 years.
- Step 4: Test rating boundaries and invalid ratings.
- Step 5: Combine service boundaries with representative ratings.
- Step 6: Verify exact bonus amount, not only the selected slab.
- Step 7: Add all discovered boundary cases to regression testing.
- Step 8: Use white-box branch/control-flow testing to confirm every decision path.
- Step 9: Record defects with expected versus actual financial outcomes.
- Step 10: Re-test after correction and perform regression on adjacent values.

11. Conclusion

The incorrect payouts at exactly 5, 10 and 15 years are strong evidence of a boundary-related weakness in the HR payroll bonus logic or its test suite. The current design is inadequate if it relies mainly on middle-of-range or random values. ECP reduces the large input space into meaningful valid and invalid classes, while BVA deliberately tests the exact decision thresholds and neighboring values. Together they provide a systematic, repeatable and auditable test design that is particularly appropriate for financial calculations. Exploratory testing remains valuable as a complementary technique, but it should not be the only protection for threshold-driven payroll logic.

12. Expected Outcome

After implementing the redesigned test suite, the system should correctly classify employees at 4, 5, 6; 9, 10, 11; and 14, 15, 16 years, correctly validate performance ratings, calculate the approved bonus amount, and maintain the same behavior after future payroll-rule changes.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements.	Clearly understands the scenario with minor gaps.	Basic understanding; some aspects of the scenario are unclear.	Poor understanding or misinterpretation of the scenario.	
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems.	Applies relevant concepts correctly with minor errors.	Applies basic concepts but with limited accuracy.	Fails to apply appropriate concepts.	
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification.	Good analysis with reasonable solutions.	Limited analysis; solutions lack justification.	Poor or no analysis; incorrect decisions.	
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach.	Logical and structured solution with minor gaps.	Basic solution with limited structure or feasibility.	Unclear or incorrect solution approach.	
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning.	Clear explanation with some justification.	Limited clarity and weak justification.	Poorly explained with no justification.	